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Children's Reading Comprehension and Metacomprehension On Screen versus On Paper

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Children's Reading Comprehension and Metacomprehension On Screen versus On Paper

Abstract

On-screen reading is becoming increasingly prevalent in educational settings, and children are now expected to comprehend texts that they read on screens. However, research suggests that reading on screen impairs comprehension compared to reading on paper. Furthermore, this medium effect is not reflected in adults' metacomprehension judgments, which often reflect greater overconfidence when reading on screen. Adults are therefore usually metacognitively unaware of the detrimental effect that on-screen reading has on their comprehension. Whether and how the medium affects children's metacomprehension has not been examined before. The main purpose of the present study was to examine the effect of the medium used for reading (screen vs. paper) on children's reading comprehension and metacomprehension. Fifth grade children ($N = 38$) read short texts, estimated their comprehension of each text, and answered a reading comprehension test. They completed this task on paper for two texts and on screen for two other texts. Results suggested that the children's reading comprehension was better when reading on paper than on screen, although initial reading time was equivalent. This paper advantage was independent of medium preferences, computer usage habits, or reading skills. Children's metacomprehension judgments were insensitive to the effect of medium, and their medium preferences further suggested that they were indifferent to the medium used for reading, both before and after experiencing the task on both media. These results suggest that children, like adults, are metacognitively unaware of the detrimental effect that on-screen reading has on their comprehension, and they are likely to make ineffective medium choices for their reading tasks.

Keywords: Reading comprehension, Metacomprehension, Children, Screen vs. paper, Reading medium

1. Introduction

Technological advancements in recent decades have brought computers into almost every aspect of our lives, and reading is no exception. Whereas traditionally reading was done predominantly on paper, on-screen reading has become increasingly prevalent, and electronic textbooks are gradually replacing printed ones (Barron, 2011; Picton, 2014). The recent inclusion of electronic assessments of reading comprehension as part of international assessments of reading skills (PIRLS, the Progress in International Reading Literacy Study, Mullis, Martin, Foy, & Hooper, 2017; PISA, the Programme for International Student Assessment, OECD, 2015) suggests that children are now expected to read and comprehend texts that are presented on screens.

Given the gradual shift to on-screen reading in education, it is essential to understand whether and how this shift affects reading comprehension. Theoretical models of reading comprehension have traditionally focused on the characteristics of the reader, the text, and the task in explaining reading comprehension (RAND Reading Study Group, 2002). In recent years, however, research suggests that reading comprehension is also affected by characteristics of text display, such as font size and type, that are unrelated to the reader, the text, or the task (e.g., Diemand-Yauman, Oppenheimer, & Vaughan, 2011; French et al., 2013; Katzir, Hershko, & Halamish, 2013; Pieger, Mengelkamp, & Bannert, 2016). It is therefore critical to examine whether the medium used for displaying the text (paper vs. screen) also plays a role in reading comprehension.

A highly related question is whether the medium used for reading affects metacomprehension. Metacomprehension is one aspect of metacognition that refers to the readers' judgments about how well they have comprehended text materials (Dunlosky & Lipko,

2007; Thomas & McDaniel, 2007; Wiley, Griffin, & Thiede, 2005; Wiley et al., 2016). Readers decide how much time and cognitive resources to invest in processing a text according to their judgment of comprehension. Therefore, when metacomprehension judgments are accurate, self-regulation of reading is usually effective, which contributes to reading comprehension

(Dunlosky, Hertzog, Kennedy, & Thiede, 2005; Thiede, Anderson, & Theriault, 2003).

Unfortunately, metacomprehension is often inaccurate. Learners, especially low ability ones, tend to overestimate their comprehension (e.g., Clinton, 2019; Hacker, Bol, Horgan, & Rakow, 2000; Maki, Shields, Wheeler, & Zacchilli, 2005; Thiede, Griffin, Wiley, & Anderson, 2010). Given the critical role of metacomprehension in reading, it is important to examine how it is affected by the medium used for reading.

The goal of the current research was to examine the effect of the medium used for reading (screen vs. paper) on both reading comprehension and metacomprehension, while focusing on a relatively unexplored age group in this respect—elementary school children.

1.1 Medium Effects on Adults

Despite the extensive use of electronic devices, most adults still prefer to read texts on paper rather than on screen, especially for learning purposes (Ackerman & Goldsmith, 2011; Baron, Calixte, & Havewala, 2017; Kurata, Ishita, Miyata, & Minami, 2017; Mizrachi, 2015; Spencer, 2006; Woody, Daniel, & Baker, 2010; but see Singer & Alexander, 2017). Is this preference justified? That is, do they benefit when reading on paper vs. on screen?

A growing body of research has examined the effect of the medium used for reading on adults' performance on reading assessment tests and on metacognition. A prominent finding is a benefit of on-paper reading over on-screen reading, both for test performance and for metacognition (e.g., Ackerman & Goldsmith, 2011; Ackerman & Lauterman, 2012; Sidi,

Shpigelman, Zalmanov, & Ackerman, 2017; Singer, Trakhman, Alexander, & Berkowitz, 2019).

In a prominent study, Ackerman and Goldsmith (2011, Experiment 2) observed that when participants were able to self-pace their reading, those who read texts on paper outperformed those who read the texts on screen on a multiple-choice test. After reading the texts but prior to taking the test, participants were asked to predict their test performance. The medium had no effect on these predictions, suggesting that readers do not appreciate the benefits of on-paper reading. Further, those predictions reflected greater overconfidence when reading on screen than when reading on paper.

Several other studies observed no effect of medium on performance or a benefit of on-screen reading (e.g., Chen & Catrambone, 2015; Daniel & Woody, 2013; Margolin, Driscoll, Toland, & Kegler, 2013; Norman & Furnes, 2016; Singer & Alexander, 2017). Importantly, however, three recent meta-analyses yielded a test performance advantage for on-paper over on-screen reading when pooling the different studies together (Clinton, 2019; Delgado, Vargas, Ackerman, & Salmerón, 2018; Kong, Seo, & Zhai, 2018) despite no difference in self-paced reading time (Clinton, 2019; Kong et al., 2018). The mixed nature of the original findings nevertheless calls for a more scrutinized examination of factors that might moderate the effect of medium. For example, the advantage of on-paper over on-screen reading was found to be greater for expository texts than for narrative texts (Clinton, 2019; Delgado et al., 2018) and when reading time was constrained rather than self-paced (Delgado et al., 2018).

The meta-analysis by Clinton (2019) also examined the effect of medium on metacognition and concluded that metacognitive assessments are more accurate when reading on paper than when reading on screen. Inaccurate metacognitive assessment is likely to yield ineffective self-regulation of cognitive effort, which could explain the inferior performance on

screen compared with on paper. Indeed, research suggests that readers tend to process information more deeply on paper than on screen (Annisette & Lafreniere, 2017; Lauterman & Ackerman, 2014; Sidi et al., 2017; Ziu, 2005).

1.2 Medium Effects on Children

A potential moderator for the effect of medium on performance is the reader's age. Children nowadays are growing up surrounded by digital technologies and spend a lot of time in front of screens (Childwise, 2017). Whereas adults usually report they prefer to read from print, preliminary findings suggest that children report a preference for on-screen reading (Dahan Golan, Barzillai, & Katzir, 2018; Huang, Liang, Su, & Chen, 2012). On-screen reading might therefore have different effects on children and adults. One view suggests that, given their early experience with technology, children are expected to achieve equivalent, or even better, comprehension levels on screen than on paper (Delgado et al., 2018; Chen, Cheng, Chang, Zheng, & Huang, 2014). Another view, in contrast, argues that children's experience with technology might encourage them to adopt speeded and shallower processing on screen (Duncan, McGeown, Griffiths, Stothard, & Dobai, 2015; Pfof, Dörfler, & Artelt, 2013), which would impair their reading comprehension.

Most of the studies that have examined the effect of medium on reading comprehension were conducted on adults. Research on children, especially elementary school children, has been much scarcer. Like the research on adults, research on children has yielded mixed results. Some studies observed better reading comprehension among children on paper than on screen (Dahan Golan et al., 2018; Eyre, Berg, Mazengarb, & Lawes, 2017; Jeong, 2012; Kerr & Symons, 2006; Lenhard, Schroeders, & Lenhard, 2017), whereas other studies yielded no effect of medium (Grimshaw, Dungworth, McKnight, & Morris, 2007; Porion, Aparicio, Megalakaki, Robert, &

Baccino, 2016; for meta-analyses, see Kingston, 2008; Wang, Jiao, Young, Brooks, & Olson, 2008) or better reading comprehension on screen than on paper (McCrea-Andrews, 2014). The recent meta-analyses that examined the effect of medium on reading comprehension found that age group (adults vs. children) did not moderate the effect (Clinton, 2019; Delgado et al., 2018). This finding should be considered with caution given the relatively small number of studies on children that were included in these meta-analyses, yet it preliminarily suggests that children, like adults, tend to comprehend texts better on paper than on screen. In a recent study by Dahan Golan and colleagues (2018), for example, fifth and sixth graders read texts and took a reading comprehension test either on paper or on screen. The results revealed that they comprehended texts better when they read on paper than when they read on screen, although the majority of children preferred to read on screen. Dahan Golan and colleagues also examined whether medium affects a specific aspect of children's metacognition, namely their confidence level in the answers they provide on the test. After answering each test question, children were asked to rate how confident they were that the answer they provided was correct. These confidence ratings were slightly higher on paper than on screen and, when compared to actual test performance, were less accurate (i.e., less calibrated) on screen than on paper. This finding suggests that medium affects children's metacognition beyond its effect on their reading comprehension.

Critically, the effect of reading medium on children's metacomprehension (i.e., self-assessment of reading comprehension after reading the text but prior to taking the comprehension test) has not been previously explored (Clinton, 2019). Given the critical role of metacomprehension for children's self-regulation of reading (de Bruin, Thiede, Camp, & Redford, 2011), the central aim of the current research was to fill this gap.

Another important question is whether there are any predictable individual differences regarding the effect of medium on children's reading comprehension. It might be, for example, that children who prefer reading on screen to reading on paper, who use computers intensively, or who have strong reading and comprehension skills would be less affected, or not affected, by the medium used for reading than children who prefer reading on paper, who use computers less often, or who have poor reading and comprehension skills. The relatively small body of research about medium effects on children's reading comprehension does not yet provide a clear answer to these questions.

1.3 The Present Study

The present study was designed to examine the effect of the medium used for reading (screen vs. paper) on children's reading comprehension and metacomprehension and to further assess whether this effect is related to children's medium preferences, computer usage habits, and reading skills. In the study, fifth grade children read short texts, estimated their comprehension of each text (i.e., metacomprehension judgment), and answered a reading comprehension test. This task was completed on paper for two texts and on screen for two other texts (within-participants). Medium preference before and after the task, computer usage habits, and reading skills were also assessed.

This design enabled us to examine the following research questions:

Question 1: Does the medium used for reading (screen vs. paper) affect children's reading comprehension? Recent evidence suggests that children comprehend texts better on paper than on screen (Dahan Golan et al., 2018), and we expected to replicate this finding. Since medium effects might depend on the technical parameters that are used for text display, a non-direct replication is important for the sake of generalization. An attempt was made in the present

study to use ecologically valid technical parameters for text display. Specifically, whereas Dahan Golan et al. presented the texts in the screen condition in a relatively large (15 point) font size, on a relatively small (15 inch) screen of a laptop computer, and in a relatively artificial manner (i.e., the screen included only the text with no buttons or any other distractions), in the present study the texts were presented using a standard (13 point) font size for this age group, on a larger (19 inch) screen of a desktop computer, and in a more ecologically valid manner (i.e., the screen also included buttons and other features that are likely to be present when reading on screen in natural settings).

Question 2: Does the medium used for reading (screen vs. paper) affect children's metacomprehension judgments? In other words, are children metacognitively sensitive to the effect of medium on their reading comprehension? This is a novel question that has not been examined before with children. Since adults' metacognitive judgments were found to be insensitive to medium effects (Ackerman & Goldsmith, 2011; Clinton, 2019), we predicted the same would be true for children.

Question 3: Do children prefer reading on paper or on screen, and does this preference change after completing a reading and comprehension task on both media? In other words, do children gain awareness to the effect of medium on their performance with experience? One recent study (Dahan Golan et al., 2018) examined this question and suggested that children prefer to read on screen but that this preference is attenuated after experiencing a reading and comprehension task on both media, and we expected to replicate these findings.

The next three questions concern individual differences in medium effects. This issue has not been investigated before. We therefore had no clear predictions and preliminarily examined these questions on an exploratory basis.

Question 4: Does the effect of medium on reading comprehension and metacomprehension depend on children's preferred medium for reading? To the extent that children's medium preferences rely on their past experience with on-screen (vs. on-paper) reading, the medium might differentially affect children who prefer reading on paper and those who prefer reading on screen (cf. Lauterman & Ackerman, 2014).

Question 5: Is the effect of medium on reading comprehension and metacomprehension related to children's computer usage habits? As suggested above (section 1.2), there are reasons to believe that higher computer usage could reduce the predicted detrimental effect of on-screen reading, but there are also reasons to believe it could enhance it.

Question 6: Is the effect of medium on reading comprehension and metacomprehension related to children's reading skills? Since on-screen reading was found to be related to shallower processing (e.g., Lauterman & Ackerman, 2014), it might have a more detrimental effect on children with less developed reading skills. Since on-screen reading was found to have detrimental effects on adults as well, however, it is equally likely to predict that medium effects will be unrelated to children's reading skills.

2. Method

2.1 Participants

Participants were 38 fifth graders ($M_{age} = 10.95$; $SD_{age} = .31$; 22 girls) from an elementary school in Israel with a population of predominantly lower- to middle-class socioeconomic background. All participants were native Hebrew speakers with no documented history of learning disabilities according to their teachers' reports. The study was approved by the Chief Scientist of the Ministry of Education in Israel as well as the Research Ethics Committee of the

School of Education at Bar-Ilan University. Before testing, written informed consent was obtained from the participants' parents.

2.2 Materials

Materials included (1) the main task that assessed reading comprehension and metacomprehension on paper and on screen, (2) a pre-task questionnaire and a post-task questionnaire that assessed medium preferences and computer usage habits, both completed on paper, and (3) two measures that assessed participants' reading skills, a single-word reading standardized test and a reading comprehension test, both completed on paper.

2.2.1 Reading comprehension and metacomprehension task

The main task assessed reading comprehension and metacomprehension on screen and on paper. It included four short age-appropriate texts¹ (103–128 words) adapted from previous national reading assessment materials and school textbooks (Katzir, Hershko, & Halamish, 2013). Materials also included four multiple-choice reading comprehension questions of various types (information finding, semantic information, syntactical information, and general inference) for each text (see Dahan Golan et al., 2018 for sample text and questions). These text questions have high reliability (Cronbach's $\alpha = .80$) according to a previous study (Katzir et al., 2013). In the current study, the Cronbach's α coefficient was .72 when calculated across the 4 texts (16 items), and .59 and .65 when calculated separately for the two sets of two texts each that were used in the paper and screen conditions (counterbalanced between participants).

The task was completed on paper for two of the texts and on a desktop computer with a 19-inch screen for the other two texts in an alternating order. The assignment of the two sets of texts to the screen and paper conditions, and whether participants completed the first text on

¹ Three of the texts were narrative and one was expository. As this text was used equally often in the screen and paper conditions across participants, genre could not account for the observed medium effects.

paper or on screen, were counterbalanced across participants. Importantly, other than the medium used for reading, the materials were presented in exactly the same manner in the paper and screen conditions (i.e., the same word documents that were used to print the materials for the paper condition were presented on the screen with 100% zoom level in the screen condition). The texts and comprehension questions were presented in the typical font size, line length, and line spacing used in national reading assessments and textbooks for fifth graders in Israel (13-pt David font, 4.6-inch line length, 1.5 line spacing; Katzir et al., 2013).

Participants read, judged their comprehension, and answered the comprehension questions for each of the four texts in turn at their own pace. For each text, participants either received a paper booklet (in the paper condition) or were presented with a word document on screen (in the screen condition) that included the text, metacomprehension judgment, and comprehension questions on separate pages. The text appeared on the first page, and no scrolling or page flipping was needed to read it. Participants were asked to read the text at their own pace and call the experimenter when they were done. The experimenter recorded their initial reading time and instructed them to turn to the next (printed or electronic) page that included the metacomprehension question. This question asked, “How well do you think you understood the text you just read?” Participants were asked to indicate their answer on a 5-point child-friendly scale developed by Kasperski and Katzir (2013), which has been used before for similar purposes (Dahan Golan et al., 2018; Halamish, Nachman, & Katzir, 2018). The scale consisted of five different facial expressions ranging from sad to happy. The faces were labeled “I didn’t understand at all” (for the saddest face), “I didn’t quite understand,” “I understood so-so,” “I understood quite well,” and “I understood very well” (for the happiest face). Responses were made by checking the appropriate label (using a pencil in the paper condition or a mouse click in

the screen condition). Upon completion, participants moved on to the next page that presented the comprehension question for that text. They answered the questions at their own pace and could go back to the text for that purpose if they wanted. Responses were made by checking the desired answer (using a pencil in the paper condition or a mouse click in the screen condition).

2.2.2 Pre-task questionnaire

The pre-task questionnaire (modified from Dahan Golan et al., 2018) assessed participants' computer usage habits and pre-task medium preferences. It included five questions: "Do you have a computer at home?" (yes/no); "Do you like to play computer games?" (yes/no); "Do you use the computer for your studies?" (yes/no); "How do you use the computer when you study?" (write assignments, search the internet for information, complete online assignments, read an electronic book, other—specify; participants were allowed to check multiple options); and "When you study, would you rather read a text on screen or on paper?" (screen/paper).

2.2.3 Post-task questionnaire

The post-task questionnaire (modified from Dahan Golan et al., 2018) assessed participants' post-task medium preferences and judgments. It included three questions: "On which medium was it easier for you to read?" (on paper/on screen/same); "On which medium do you think you performed better?" (on paper/on screen/same); "If you were to read another text now, on which medium would you prefer to read it?" (on paper/on screen/doesn't matter).

2.2.4 Single word reading test

Reading skill in terms of reading speed was assessed using a single-word reading standardized test (Katzir, Schiff, & Kim, 2012; Schiff, Kahta, & Katzir, 2006). In this test, participants (tested individually) were presented with 78 real words of increasing difficulty, arranged in four columns on a sheet of paper. They were asked to read aloud as many words as

possible in 45 seconds. The experimenter documented the number of words that were accurately read and stopped the child after 45 seconds. Scores reflected the number of words accurately read in that time.

2.2.5 Baseline reading comprehension test

Reading skills in terms of comprehension were assessed with a baseline reading comprehension test. In this test, participants were asked to read a 303-word expository text and to answer four multiple choice comprehension questions on it. The score reflected the number of correct answers. The task was performed on paper. The Cronbach's alpha of this test in the current study was .57.

2.3 Procedure

The single word reading test was completed individually. All other tasks were completed in groups of 10 to 20 children each, and the experimenter was always present. The study was conducted in two sessions on different days. During the first session, participants answered the pre-task questionnaire and completed the single-word reading test and the baseline reading comprehension test. During the second session, participants completed the main reading comprehension and metacomprehension task and answered the post-task questionnaire.

2.4 Data analysis

First, the main dependent measures were calculated. A reading comprehension score was computed for each participant and medium by calculating the proportion of correct reading comprehension questions across the two texts. The metacomprehension judgments were coded on a scale of 1 (lowest judgment) to 5 (highest judgment), and then mean metacomprehension judgments were computed for each participant and medium by averaging the metacomprehension judgments across the two texts. Finally, medium effect scores on

comprehension and on metacomprehension were computed for each participant by subtracting the score on screen from the score on paper.

Then, data were quantitatively analyzed to assess each research question, using IBM SPSS Statistics version 25 (IBM Corp, 2017). To answer research questions 1 and 2, paired-sample *t*-tests were conducted on the reading comprehension scores and on the mean metacomprehension judgments (respectively), with medium as the independent variable. In a follow-up analysis, comprehension scores and metacomprehension judgments were analyzed together in a two-way repeated measures ANOVA, with medium (screen, paper) and measure (comprehension scores, metacomprehension judgments) as the independent, repeated measures. To answer research question 3, pre-task and post-task preferences and judgments were analyzed using chi-squared tests. To answer research questions 4, 5 and 6, correlations were computed between the medium effect scores on comprehension and on metacomprehension, on the one hand, and medium preferences, computer usage habits, and reading skills (respectively), on the other hand.

Sensitivity analyses were conducted using G*Power version 3.1.9.2 (Faul, Erdfelder, Buchner, & Lang, 2009). With an alpha level of .05 and .80 power, the sample size ($N = 38$) afforded sufficient sensitivity to detect moderate effects ($d \geq .41$ and $d \geq .47$ for one-tailed and two-tailed *t* tests, respectively; $w \geq .50$ for goodness-of-fit chi squared tests; $\rho \geq .42$ and $\rho \geq .44$ for two-tailed point-biserial and Pearson correlations, respectively).

3. Results

Scores on the standardized single-word reading test suggested that all children were in the normal age-appropriate range according to local norms. All children were therefore included in the analyses. Results are presented below by the corresponding research questions.

3.1. Question 1: Does the medium used for reading (screen vs. paper) affect children's reading comprehension?

Mean reading comprehension scores by medium are presented in Figure 1. A paired-sample t -test suggested that reading comprehension scores were significantly higher when reading on paper ($M = .62$, $SD = .23$) than when reading on screen ($M = .52$, $SD = .27$), $t(37) = 2.17$, $p = .036$ (two-tailed; $p = .018$ one-tailed), Cohen's $d = .41$. Initial reading time was computed for each participant and medium by calculating the average initial reading time (in seconds) across the two texts. A paired-sample t -test suggested that there was no significant difference in initial reading time between the paper ($M = 42$, $SD = 38$) and screen ($M = 43$, $SD = 29$) conditions, $t(37) = .35$, $p = .729$ (two-tailed), Cohen's $d = .03$. Therefore, reading on paper not only led to superior comprehension but was also more efficient than reading on screen because this superior comprehension was achieved without a cost in reading time.

3.2 Question 2: Does the medium used for reading (screen vs. paper) affect children's metacomprehension judgments?

Mean metacomprehension judgments by medium are presented in Figure 2. A paired-sample t -test suggested that metacomprehension judgments were numerically higher in the screen condition ($M = 4.42$, $SD = .56$) than in the paper condition ($M = 4.29$, $SD = .70$), but this difference was not statistically significant, $t(37) = 1.38$, $p = .177$ (two-tailed), Cohen's $d = .21$. Children's metacomprehension judgments were therefore insensitive to the advantage of paper over screen for comprehension.

The null effect of medium on metacomprehension judgments is especially interesting in light of the significant effect of medium on comprehension scores that was reported in section 3.1. We therefore followed up on the separate analyses of comprehension scores and

metacomprehension judgments with an analysis that included both measures. Since the metacomprehension judgments were made on a 1–5 scale that was different from the 0–1 scale used to assess reading comprehension scores, they were first converted to a 0–1 scale (by subtracting 1 from the mean rating and dividing the difference by 4). Then, a two-way repeated measures ANOVA was conducted, with medium (screen, paper) and measure (comprehension scores, metacomprehension judgments) as the independent, repeated measures. This analysis yielded a significant main effect of measure, $F(1, 37) = 48.30, p < .001, \eta_p^2 = .57$. This effect is difficult to interpret because originally, the comprehension and metacomprehension scores were not measured on the same scale (Dunlosky, Mueller, & Thiede, 2016). The main effect of medium was not significant, $F(1, 37) = 1.89, p = .177, \eta_p^2 = .05$. Importantly, however, the interaction between medium and measure was significant, $F(1, 37) = 6.23, p = .017, \eta_p^2 = .14$. This interaction suggests that indeed, medium affected comprehension scores (better comprehension on paper than on screen) but metacomprehension judgments were insensitive to this effect.

3.3 Question 3: Do children prefer reading on paper or on screen, and does this preference change after completing reading and comprehension tasks on both media?

Prior to completing the main task, the proportion of participants who reported that they prefer reading on screen (58%) was numerically higher than the proportion of participants who reported that they prefer reading on paper (42%), but these proportions were not significantly different from each other, $\chi^2(1) = .95, p = .330$. Participants therefore had no significant prior preference for reading on paper vs. on screen.

In the post-task questionnaire, participants were asked to compare their performance in the paper and screen conditions. In response to the question of which medium was easier for

them to read, 37% of the participants responded “on paper,” 24% responded “on screen,” and 40% responded “the same.” A chi-squared test suggested that these proportions were not significantly different from each other, $\chi^2(2) = 1.63, p = .442$. In response to the question of in which medium they thought they performed better, 24% of the participants responded “on paper,” 24% responded “on screen,” and 52% responded “the same.” A chi-squared test revealed a significant difference between these proportions, $\chi^2(2) = 6.37, p = .041$, suggesting that the proportion of participants who responded “the same” was larger than the other two proportions. Participants were also asked in the post-task questionnaire to indicate their medium preference if they were to read another text. Responses suggested that 24% of the participants preferred to read another text on paper, 24% preferred to read another text on screen, and 52% had no preference. A chi-squared test revealed a significant difference between these proportions, $\chi^2(2) = 6.37, p = .041$, suggesting that the proportion of participants who had no clear post-task preference was higher than the other two proportions.

In summary, responses to the post-task questionnaire reveal that, even after completing the reading comprehension and metacomprehension tasks on both paper and screen, and despite a clear paper advantage for reading comprehension, participants were still unable to appreciate this effect and had no clear medium preference for future tasks.

3.4 Question 4: Does the effect of medium on reading comprehension and metacomprehension depend on children’s preferred medium for reading?

Did the effect of medium on reading comprehension and metacomprehension depend on participants’ prior medium preference? To answer this question (as well as the following ones), medium effect scores on comprehension and metacomprehension were computed for each participant by subtracting the score on screen from the score on paper. The mean medium effect

on comprehension score was positive ($M = .10$, $SD = .30$), reflecting the fact that the comprehension scores on paper were higher than the comprehension scores on screen. The mean medium effect on metacomprehension score was near zero ($M = -.03$, $SD = .15$), reflecting the fact that the metacomprehension judgments on paper and on screen were similar.

The relationships between these medium effect scores and medium preference (coded as 0–computer, 1–paper) were then examined. For reading comprehension, a point-biserial correlation suggested that there was no significant relationship between medium preference and medium effect, $r = -.15$, $p = .380$. To illustrate, Figure 3 presents the distribution of participants by medium preference and medium effect on comprehension categorized by those who comprehended better on paper, those who comprehended the same on screen and on paper, and those who comprehended better on screen. As the figure indicates, the percentage of participants who comprehended better on paper was higher than the percentage of participants who comprehended better on screen, regardless of whether they initially preferred to read on paper (63% vs. 31%, respectively) or on screen (73% vs. 27%, respectively). A chi-squared test suggested that there was no relationship between medium preferences and categorized medium effect on comprehension, $\chi^2(2) = 1.57$, $p = .457$.

For metacomprehension, a point-biserial correlation suggested that there was no significant relationship between medium preference and medium effect, $r = .10$, $p = .545$. To illustrate, Figure 4 presents the distribution of participants by medium preference and medium effect on metacomprehension categorized by those whose mean metacomprehension judgments were higher on paper, those whose mean metacomprehension judgments were the same on screen and on paper, and those whose mean metacomprehension judgments were higher on

screen. A chi-squared test suggested that there was no relationship between medium preferences and categorized medium effect on metacomprehension, $\chi^2(2) = 2.28, p = .320$.

In summary, these analyses converge to suggest that children had no clear prior medium preference and that the effect of medium on reading comprehension and metacomprehension was unrelated to participants' medium preferences. Comprehension was better on paper than on screen, and participants were metacognitively insensitive to this effect, regardless of their prior medium preferences.

3.5 Question 5: Is the effect of medium on reading comprehension and metacomprehension related to children's computer usage habits?

Overall, 97% of the participants reported having a computer at home, 71% reported that they like to play computer games, and 63% reported that they tend to use the computer for their studies. On average, participants checked 2.13 ($SD = 1.37$; range = 0–4) out of the five different options that were provided for using the computer when studying.

Are computer usage habits related to the effect of medium on reading comprehension and metacomprehension? To answer this question, we computed a computer usage habits score by summing up the responses to the four computer usage habits questions from the pre-task questionnaire.² The mean computer usage habits score was 2.75 (on a scale of 0–4; $SD = .99$; range = 0–3.80). We then examined the relationship between computer usage habits scores on one hand, and medium effect scores on comprehension and metacomprehension on the other hand. The Pearson correlation between computer usage habits and medium effect on comprehension was not statistically significant, $r = -.21, p = .199$. The Pearson correlation

² A positive answer to each of the first three questions (Do you have a computer at home?; Do you like to play computer games?; Do you use the computer for learning?) contributed one point to the total score, and up to one more point was added based on the proportion of the computer usage habits checked out of five in the fourth question (e.g., the computer usage score for participants who answered positively on the first three questions and marked two options out of five in the fourth question was 3.4).

between computer usage habits and medium effect on metacomprehension was not statistically significant either, $r = .04$, $p = .797$. Therefore, the results suggest that the effect of medium on reading comprehension and metacomprehension was unrelated to children's computer usage habits.

3.6 Question 6: Is the effect of medium on reading comprehension and metacomprehension related to children's reading skills?

To answer this question, we examined the relationship between reading speed (as assessed by the score in the single-word reading test) and reading comprehension (as assessed by the score in the baseline reading comprehension test) on one hand, and medium effects on comprehension and metacomprehension on the other hand. The results suggested that the medium effect on comprehension was not significantly related to reading speed ($r < .01$, $p = .994$) or reading comprehension ($r = .02$, $p = .924$). Likewise, the medium effect on metacomprehension was not significantly related to reading speed ($r = -.01$, $p = .972$) or reading comprehension ($r = -.07$, $p = .689$). The effect of medium on reading comprehension and metacomprehension was therefore unrelated to children's reading skills.

4. Discussion

The results of the present study suggest that elementary school children understand texts better when reading them on paper than on screen. These results are consistent with previous findings (Dahan Golan et al., 2018) and further generalize and extend them. The present study generalizes the previous results by using a different sample and more ecologically valid technical parameters for text display in terms of font size, screen size, and on-screen distractions. The present study further extends the previous results by pointing to the robustness of the effect of medium on reading comprehension. It suggests that the advantage of on-paper over on-screen

reading is unrelated to children's initial reading preference, computer usage habits, or reading ability.

A unique aspect of the current research was that it examined the effect of medium not only on children's reading comprehension but also on their metacomprehension. Results clearly suggest that children are metacognitively unaware of the effect of medium on their comprehension. They comprehend texts better on paper than on screen, but they do not know that and judge their comprehension, after reading the texts and before seeing the test, to be grossly the same on screen and on paper.

Another interesting finding of the present research is that children had no prior (significant) preference for reading on paper or reading on screen (although numerically, the majority preferred reading on paper) and, importantly, still had no clear preference for future reading even after completing the main task both on paper and on screen. Thus, the preference data converge with the metacomprehension judgments to suggest that children do not appreciate the disadvantage that on-screen reading has on their reading comprehension.

Some limitations of the present study should be mentioned. First, the current design did not allow us to examine whether the effect of medium on reading comprehension and metacomprehension depends on text genre. Given that meta-analyses conducted with studies that examined mainly adults suggested that the effect of medium on reading comprehension tends to be larger for expository than for narrative texts (Clinton, 2019; Delgao et al., 2018), the potential moderating role of text genre among children remains an important issue for future research. Second, the results of the present study suggest that the effect of medium was unrelated to reading skills, but reading skills were only preliminarily assessed using two simple and short tasks (a single-word reading test and a short reading comprehension test). The potential role of

reading skills on medium effects should be further examined with a better profiling of reading skills. Third, the metacomprehension judgments were measured on a simplistic, 5-point ordinal scale, and this scale was incomparable to the scale that was used to assess reading comprehension (proportion of correct reading comprehension questions). Therefore the accuracy of these judgments (in terms of calibration, which reflects over/under confidence) could not have been determined (Dunlosky, Mueller, & Thiede, 2016). Using comparable scales in the future would allow one to compute judgments' accuracy and examine whether it is affected by medium. Finally, the present study examined children's reading comprehension and metacomprehension when reading time was self-paced. Delgado and colleagues (2018), however, found that the advantage of on-paper reading increases when reading time is limited (compared to self-paced). Future research should examine medium effects among children under fixed (experimenter-paced) conditions and limited reading time.

The current results have important educational implications. On-screen reading tasks are becoming increasingly prevalent in schools. The finding that elementary school children comprehend text better on paper than on screen suggests that the growing transition to on-screen reading in educational settings might be premature. The finding that children are metacognitively unaware of the detrimental effect of on-screen reading and have no clear medium preference further suggests that children are likely to make ineffective medium choices for their reading tasks. It might therefore be useful to share the current results with children and their teachers to help them make better medium choices to meet their reading goals. Finally, the current results point to the importance of the future development of interventions that could help children to overcome the detrimental effect of on-screen reading and improve their comprehension when reading on screens.

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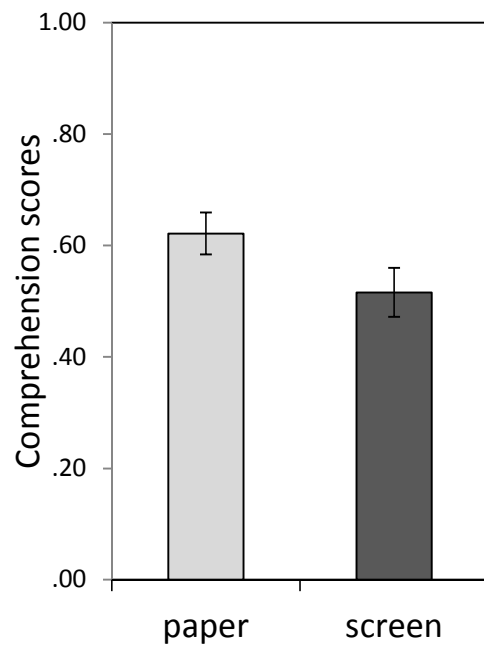


Figure 1. Mean reading comprehension score by medium. Error bars represent +/-1 standard error of the mean.

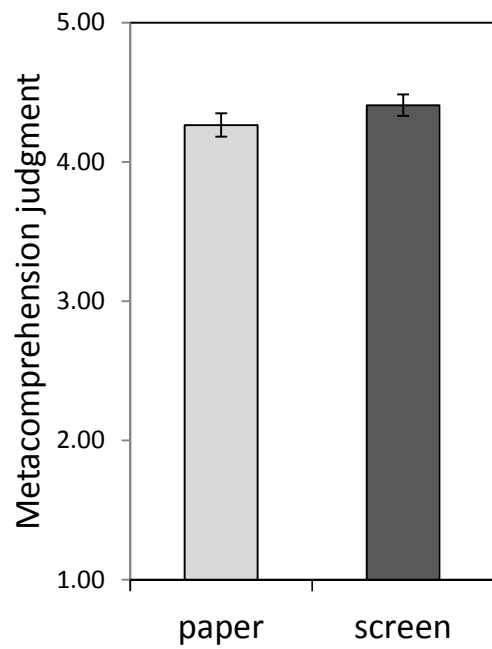


Figure 2. Mean metacomprehension judgment by medium. Error bars represent +/-1 standard error of the mean.

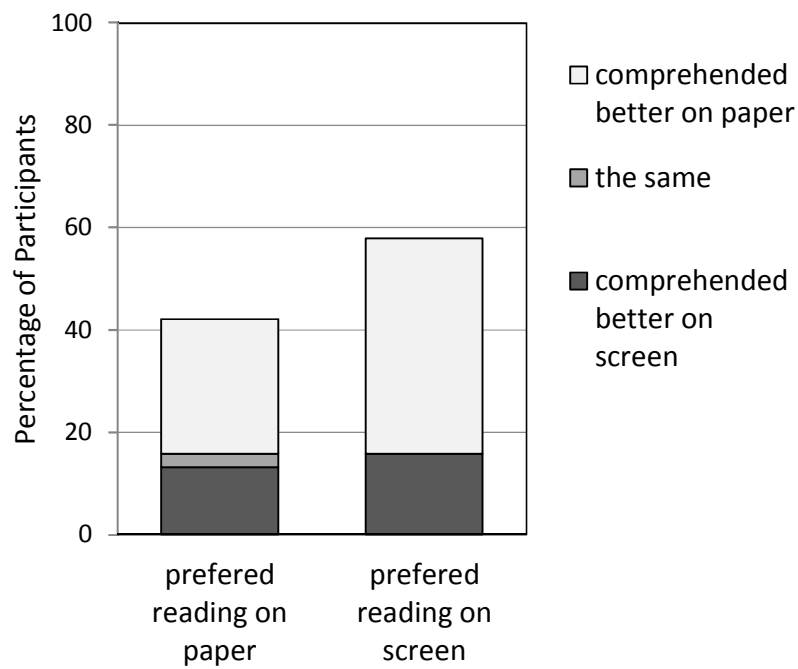


Figure 3. Distribution of participants by relative comprehension on paper vs. on screen and pre-task medium preferences. The Y-axis depicts the percentage of participants for each category out of the total number of participants in the sample.

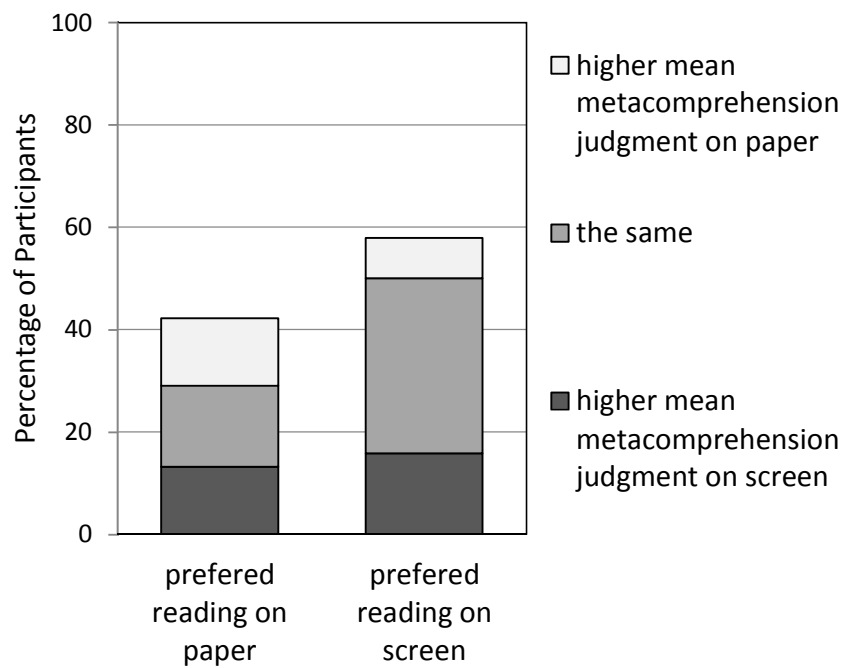


Figure 4. Distribution of participants by relative metacomprehension on paper vs. on screen and pre-task medium preferences. The Y-axis depicts the percentage of participants for each category out of the total number of participants in the sample.

Highlights

- Reading on screen impaired children's comprehension compared to reading on paper
- Their metacomprehension judgments suggested that they were unaware of this effect
- Children had no clear a-priori preference for reading on paper versus on screen
- They were insensitive to the effect of medium even after experiencing it
- The effect of medium was unrelated to preferences, computer usage or reading skills